

PHYSICS CONTRIBUTION

Prospective Clinical Validation of Virtual Patient-Specific Quality Assurance of Volumetric Modulated Arc Therapy Radiation Therapy Plans



Phillip D.H. Wall, PhD, MS, Emily Hirata, PhD, Olivier Morin, PhD, Gilmer Valdes, PhD, and Alon Witztum, DPhil

Department of Radiation Oncology, University of California, San Francisco, California

Received Jul 20, 2021; Accepted for publication Apr 27, 2022

Purpose: Performing measurement-based patient-specific quality assurance (PSQA) is recognized as a resource-intensive and time inefficient task in the radiation therapy treatment workflow. Paired with technological refinements in modern radiation therapy, research toward measurement-free PSQA has seen increased interest during the past 5 years. However, these efforts have not been clinically implemented or prospectively validated in the United States. We propose a virtual QA (VQA) system and workflow to assess the safety and workload reduction of measurement-free PSQA.

Methods: An XGBoost machine learning model was designed to predict PSQA outcomes of volumetric modulated arc therapy plans, represented as percent differences between the measured ion chamber point dose in a phantom and the corresponding planned dose. The final model was deployed within a web application to predict PSQA outcomes of clinical plans within an existing clinical workflow. The application also displays relevant feature importance and plan-specific distribution analyses relative to database plans for documentation and to aid physicist interpretation and evaluation. VQA predictions were prospectively validated over 3 months of measurements at our clinic to assess safety and efficiency gains.

Results: Over 3 months, VQA predictions for 445 volumetric modulated arc therapy plans were prospectively validated at our institution. VQA predictions for these plans had a mean absolute error of $1.08\% \pm 0.77\%$, with a maximum absolute error of 2.98%. Using a 1% prediction threshold (ie, plans predicted to have an absolute error $<1\%$ would not require a measurement) would yield a 69.2% reduction in QA workload, saving 32.5 hours per month on average, with 81.5% sensitivity, 72.4% specificity, and an area under the curve of 0.81 at a 3% clinical threshold and 100% sensitivity, 70% specificity, and an area under the curve of 0.93 at a 4% clinical threshold.

Conclusions: This is the first prospective clinical implementation and validation of VQA in the United States, which we observed to be efficient. Using a conservative threshold, VQA can substantially reduce the number of required measurements for PSQA, leading to more effective allocation of clinical resources. © 2022 Elsevier Inc. All rights reserved.

Corresponding author: Phillip D.H. Wall, PhD, MS; E-mail: phillipdhwall@gmail.com

Disclosures: A.W. and G.V. report an ownership stake in Foretell Med, LLC.

Data sharing statement: Research data are stored in an institutional repository and will be shared upon request to the corresponding author.

Acknowledgments—We acknowledge Tomi Nano, Matthieu Lafreniere, Dianne Ferguson, Annette Villa, and Devan Krishnamurthy for their assistance with data acquisition.

Introduction

Volumetric modulated arc therapy (VMAT) was the culmination of decades of technological innovations and has become an essential treatment modality in radiation oncology centers globally.^{1,2} As a rotational form of intensity modulated radiation therapy (IMRT), VMAT yields significant improvements in target coverage, sparing of surrounding healthy tissues, and treatment efficiency.³⁻⁷

With these clinical advantages, there are additional quality assurance (QA) considerations. VMAT treatments require the linear accelerator (linac) to continually synchronize its gantry rotation, multileaf collimator (MLC) movements, and dose rate variations, necessitating additional performance tests to verify the accurate delivery of this type of treatment.^{8,9} Inverse optimization facilitates the efficient design of VMAT treatment plans for each patient, but this trial-and-error process has contributed to variations in plan quality.¹⁰ Additionally, the VMAT technique often results in low dose baths to all organs near the treated area and increased plan complexity, especially for hypofractionated treatment schemes such as SBRT and SRS. Plan complexity can be described by increased beam modulation, number of monitor units (MUs), uncertainties in small field dosimetry due to loss of charged particle equilibrium, and susceptibility to interplay between movements of linac and internal organs.¹¹⁻¹⁴ Furthermore, these VMAT plans can result in numerous irregularly shaped and/or small beam segments to obtain improved target coverage and sparing of healthy tissues. Small beam segments introduce higher degrees of dosimetric uncertainty, placing an emphasis on the treatment planning system's (TPS) ability to accurately model lateral electron scatter, MLC leaf ends, leaf transmission, and interleaf leakage.¹⁵ Collectively, these hardware and software complexities require a robust and efficient QA program that ensures VMAT treatment plans are delivered as intended and verifies the accuracy of the TPS dose calculation and beam model.

The standard-of-care of VMAT QA programs includes a pretreatment patient-specific measurement, also referred to as patient-specific quality assurance (PSQA). The process typically involves copying the approved VMAT beams from the patient's anatomy to a phantom with known geometry, and recomputing the dose. The recomputed plan is delivered to the phantom housing one or more radiation detectors to measure the dose, which is then compared with the calculated dose to determine whether the plan is approved for treatment. Although these pretreatment measurements serve as the basis of most modern VMAT QA programs, many different specific implementations exist.¹⁶ Additionally, several software-based methods have been recently developed that can increase the robustness of any measurement-based QA program.¹⁷⁻¹⁹

Measurement-based PSQA of VMAT treatment plans is a time- and resource-intensive task for radiation oncology departments across the globe in terms of equipment,

personnel, and workload. A 2019 survey of more than 400 clinics across China revealed difficulties in securing the necessary resources (eg, physicist time) to rigorously implement and perform VMAT PSQA, which likely reflects similar challenges throughout the world.²⁰ Additionally, the value and effectiveness of pretreatment PSQA has come under increasing scrutiny in terms of error-detection, where Ford et al²¹ found PSQA to be the least effective out of 15 individual quality control checks for detecting reported high severity incidents.²¹⁻²³ A recent report by the Imaging and Radiation Oncology Core suggests independent recalculation software, such as Mobius3D (M3D; Varian Medical Systems, Inc, Palo Alto, CA), outperforms many measurement-based IMRT PSQA methods at detecting unacceptable plans.¹⁸ Furthermore, replans due to a failed PSQA measurement is disruptive to conventional clinical workflows, often creating confusion and frustration for patients. Given these questions of effectiveness along with the consistent refinement of modern linac and TPS technologies, numerous studies have explored measurement-free IMRT and VMAT QA frameworks to maximize efficiency without compromising patient safety using various predictive methods,²⁴⁻³⁶ with sentiments emerging that measurement-free PSQA is ready for clinical practice.³⁷

Our group has been active in this space of applying machine learning (ML) models for predicting PSQA outcomes based on complexity features extracted from individual treatment plans. First, a generalized Poisson regression model with Lasso regularization was trained and tested using 78 plan complexity features and found to predict gamma passing rates with 3% accuracy for fixed-gantry IMRT plans.²⁹ This seminal work demonstrated the feasibility of robustly and safely monitoring IMRT plans without requiring measurements. Additional studies have surveyed and tested other machine learning algorithms for predicting gamma passing rates of VMAT plans with similar accuracy.³⁰ All of these previous studies were retrospective, and predicted QA outcomes based on measurements from a wide range of devices that typically present disadvantages in terms of uncertainties or dependencies (eg, diode-based arrays), or analysis shortfalls (eg, gamma analysis²²) compared with ion chamber measurements, considered among the most accurate and reliable for absolute dose measurements.

The emergence of the COVID-19 pandemic has forced sweeping shifts in working environments and staffing models in almost every industry, including radiation oncology. Our department spans 3 distinct clinical facilities sprawled across one of the most densely populated large cities in the United States, and moving to a hybrid on-site and remote working environment presented challenges of ensuring safe patient treatments while minimizing COVID-19 exposure risk for on-site team members. This served as the motivation for our proposal of a ML model paired with an independent dose calculation system that would reduce the number of dose measurements for our QA team.³⁸ Although

other software-based PSQA methods could help minimize COVID-19 exposure risk as well, a predictive model provides and maintains a robust relationship with established measurement-based PSQA outcomes that other methods cannot as of yet. Furthermore, virtual QA (VQA) studies have yet to demonstrate successful and effective clinical implementation of such a model, let alone on a large prospective validation data set.

To our knowledge, this is the first model based solely on plan parameters to predict ion chamber QA measurements. In this study, we evaluate a ML model for predicting ion chamber disagreements for VMAT PSQA and demonstrate the first prospective clinical implementation and validation of a VQA model in the United States at a multisite institution.

Methods and Materials

Our PSQA program uses M3D as an independent dose calculation verification paired with a point dose ion chamber (IC) measurement in a water-equivalent Mobius QA phantom (23 cm × 26 cm × 10 cm) with 7 possible chamber insert positions. M3D performs a dose recalculation using a collapsed cone superposition algorithm for the approved patient treatment plan and compares its results against the TPS calculations in the form of dose volume histograms and 3-dimensional (3D) gamma analyses.

An appropriate IC measurement position is selected for each VMAT plan in the TPS, ideally in a high-dose, low-gradient region. Specifically, the active volume of the IC must lie within a region where the average computed dose exceeds 80% of the prescribed fractional dose with a standard deviation of less than 5%. It is important to note that an acceptable measurement region may not necessarily exist at the 7 possible chamber insert positions. In this case, the phantom is shifted (via translations of the treatment couch upon which the phantom rests) such that one of the 7 chamber insert positions intersects an acceptable measurement region. The plan is delivered and an absolute point dose measurement is collected with a small volume IC (PTW 31022 PinPoint 3D; 0.016-cm³ volume), cross-calibrated according to clinical reference dosimetry standards, that is, AAPM TG-51³⁹ with an ion chamber traceable to an Accredited Dosimetry Calibration Laboratory, placed in the appropriate phantom insert. The percent difference between the measured dose and the TPS dose is calculated and logged in an institutional REDCap database. The clinical action thresholds for these IC disagreements are 3% and 5% for SBRT and non-SBRT IMRT plans, respectively.

VQA model development

Data

A data set of 579 VMAT treatment plans and associated IC measurements from 2018 to 2020 was retrospectively collected at our institution. Among these plans, 457 plans were

delivered on 1 of 3 matched Varian TrueBeam linacs (Varian Medical Systems) at site A, 103 plans were delivered on a Varian TrueBeam STx at site B, and 19 plans were delivered on an Elekta Versa HD at site C. All plans were generated using Pinnacle version 16.0 (Philips Radiation Oncology Systems, Fitchburg, WI) or RayStation version 7 (RaySearch Laboratories, Stockholm, Sweden) TPS.

In-house software was developed to extract 30 classes of complexity features representing different failure modes (totaling 293 raw features) from each VMAT DICOM RT-Plan file (Table 1).^{29,40-44} These complexity features are commonly used in the literature and have been found to be useful for predicting QA outcomes.^{25,34-38} Three features were classified as categorical, while the rest were numerical. QA outcomes, defined as the percent difference between planned dose and ion chamber measured dose, were extracted from the secure REDCap database used to log PSQA measurements. The mean absolute error of these 579 ion chamber measurements was 1.36% with a standard deviation of 0.96%.

Model training and testing

An XGBoost model was employed in this study to predict IC chamber disagreements using open-source software.⁴⁵ The XGBoost algorithm was chosen for its flexibility, scalability, built-in feature importance analysis and selection, and demonstrated performance capabilities with high-dimensional data sets.⁴⁶ Before model training, the extracted complexity features were processed, whereby numerical features were standardized such that each feature had a mean value of zero and standard deviation of one and the categorical features were one-hot encoded. Note that QA outcomes queried from the clinical database for model development were assumed to have been acquired within an acceptable high-dose, low-gradient measurement region. Therefore, model predictions reflect ion chamber disagreements assuming clinically acceptable measurement conditions. Furthermore, model results reflect outcome predictions for a representative clinical PSQA measurement, but not one specific measurement point within the QA phantom. A 10-fold cross-validation combined with a grid search was used to tune the maximum tree depth (values ranging from 1-9) and learning rate (values ranging from 0.01-0.5) hyperparameters, after which the tuned model was trained and tested on a randomized 90/10 data set split, respectively. After the tuned model was evaluated, the entire data set was used to train the model before deployment into our clinical workflow for prospective clinical validation. Receiver operator characteristic (ROC) analysis was performed with this tuned model upon the entire retrospective data set to outline baseline model performance and to assess potential clinical decision thresholds for desired levels of sensitivity and specificity. Although the underlying predictive model was designed for a regression task (ie, generating continuous ion chamber measurement disagreement predictions), its viability as a classification decision-support tool was the focus of this study given previous evidence and preliminary investigations showing weak regression performances. This decision to use a regression prediction model was based on

Table 1 Summary of complexity feature groups

Feature group	Primary failure mode	Reference(s)	Notes
MCS	CPE/small field dosimetry/TPS dose calculation failure	McNiven et al ⁴³ and Masi et al ⁴²	
EM	Tongue and groove effect	Younge et al ⁴⁴	
LT	MLC leaf transmission		Total leaf travel per leaf; includes first, second, third, fourth, and fifth moments unweighted and weighted by segment MU
LM	MLC leaf transmission		Defined as average leaf travel per degree of gantry rotation; includes first, second, third, fourth, and fifth moments unweighted and weighted by segment MU
Arc length	CPE/small field dosimetry/TPS dose calculation failure		Total degrees of gantry rotation in plan
MU factor	CPE/small field dosimetry/TPS dose calculation failure		Total planned MUs normalized by fractional dose to specification point
Number of arcs/beams	CPE/small field dosimetry/TPS dose calculation failure		
Average JP	Jaw transmission		Feature for each jaw and feature with average combined jaw position
JT	Jaw transmission		Total travel for each jaw; weighted and unweighted
JM	Jaw transmission		Average jaw travel per degree of gantry rotation for each jaw; unweighted and weighted
Average collimator angle	CPE/small field dosimetry/TPS dose calculation failure		
AA	MLC leaf transmission	Du et al ⁴¹	MLC aperture area (equivalent to CIAO); total along with first, second, third, fourth, and fifth moments unweighted and weighted by segment MU
AP	MLC leaf transmission	Du et al ⁴¹	MLC aperture perimeter; total along with first, second, third, fourth, and fifth moments unweighted and weighted by segment MU
PI	MLC leaf transmission	Du et al ⁴¹	MLC Plan Irregularity; total along with first, second, third, fourth, and fifth moments unweighted and weighted by segment MU
LG	Leaf end leakage		Distance between opposing in-field leaf pairs; first, second, third, fourth, and fifth moments unweighted and weighted by segment MU
PM	MLC leaf transmission	Du et al ⁴¹	

(Continued)

Table 1 (Continued)

Feature group	Primary failure mode	Reference(s)	Notes
SAS	CPE/small field dosimetry/TPS dose calculation failure	Crowe et al ⁴⁰	Additional feature included maximum SAS among all control points (2, 5, 10, 11, 12, 13, 14, 15, 20, 25, . . . , 125 mm)
Closed leaf score	MLC leaf transmission	Crowe et al ⁴⁰	Not included in previous work since it is always 0
CAS	Tongue and groove effect	Crowe et al ⁴⁰	
FAOC	CPE/small field dosimetry/TPS dose calculation failure	Valdes et al ²⁹	First, second, third, fourth, and fifth moments unweighted and weighted by segment MU (5, 10, . . . , 40 mm)
Nominal dose rate	CPE/small field dosimetry/TPS dose calculation failure		
Treatment machine	CPE/small field dosimetry/TPS dose calculation failure		One of 4 dosimetrically matched machines; categorical
FFF	CPE/small field dosimetry/TPS dose calculation failure		Categorical
MLC head	Tongue and groove effect		Categorical
JAA	Jaw transmission	Du et al ⁴¹	Total along with first, second, third, fourth, and fifth moments unweighted and weighted by segment MU
AP_jaw	Jaw transmission	Du et al ⁴¹	Total along with first, second, third, fourth, and fifth moments unweighted and weighted by segment MU
JPI	Jaw transmission	Du et al ⁴¹	Total along with first, second, third, fourth, and fifth moments unweighted and weighted by segment MU
Aperture ratio	MLC leaf transmission	Valdes et al ²⁹	AA/JAA; average weighted and unweighted
Fraction of MLC area less than square	MLC leaf transmission	Valdes et al ²⁹	10, 20, 30, 40, 50 mm

Abbreviations: AA = aperture area; AP = aperture perimeter; AP_jaw = jaw aperture perimeter; CAS = cross-axis score; CPE = charged particle equilibrium; EM = edge metric; FAOC = fractional area outside of circle; FFF = flattening filter free; JAA = jaw aperture area; JPI = jaw plan irregularity; JP = jaw position; JM = average jaw motion; JT = jaw travel; LG = leaf gap; LM = leaf motion; LT = leaf travel; MCS = modulation complexity score; PI = plan irregularity; SAS = small aperture score.

the intent of using the raw data. Although the PSQA result of ultimate clinical interest is binary (ie, pass or fail), a plan that passes with an elevated error rate should glean more insight from a failure mode perspective compared with one that passes with less error. As such, we made the initial decision to investigate this PSQA problem with regression to include all the information available, while analyzing its potential clinical value as a classification decision-support supplement.

Prospective clinical validation

A web-based VQA application was developed using the open-source library Streamlit⁴⁷ to enable prospective predictions of IC chamber measurements and comparative analysis of the resulting output from the XGBoost model (Fig. 1). The application provides a searchable database view plans available for review. For a selected plan, the application displays relevant plan-specific feature importance analysis relative to database plans (used to train the predictive XGBoost model) for physicist evaluation and documentation. Specifically, the VQA application defaults to display the 5 most important features for the model by total gain along with brief descriptors and failure modes of each feature. Accompanying this feature importance summary is a boxplot illustrating the distributions of the numerical features found in the model data set, along with feature values of the predicted plan relative to these distributions. The final model was implemented within this web application to prospectively make predictions within the normal clinical workflow.

This VQA application requires no additional button clicks in the clinical planning workflow to export the plan to the

VQA database, as this DICOM RT-Plan export was added to the existing clinical script used by the dosimetry team to generate the initial QA plan. The DICOM RT-Plan files are exported to a folder on the local hospital network, where an independent Python watchdog application is triggered to parse the file, extract the necessary metadata and complexity features, and call the XGBoost model to calculate an IC measurement prediction. The process takes less than 10 seconds per plan on average, running on a machine with a 3 GHz Intel Core 2 Duo processor and 8 GB of RAM. All data are stored in a VQA SQL database for the web application to query on user interactions. VQA predictions were prospectively recorded alongside clinical PSQA measurements. Additionally, to estimate any potential time savings associated with measurement-free VQA, the clinical QA team recorded the approximate overall time they spent performing PSQA each day (including phantom setup) along with the number of plans measured each day across all sites over the course of 1 month. The average time spent per plan during this period was used to assess any potential time saving associated with VQA implementation in the absence of PSQA measurements. Prediction error was evaluated and ROC analysis was performed on this prospective data set as well.

Results

VQA model development

Table 2 reports the ten most important features as determined by the XGBoost model in terms of coverage, gain,

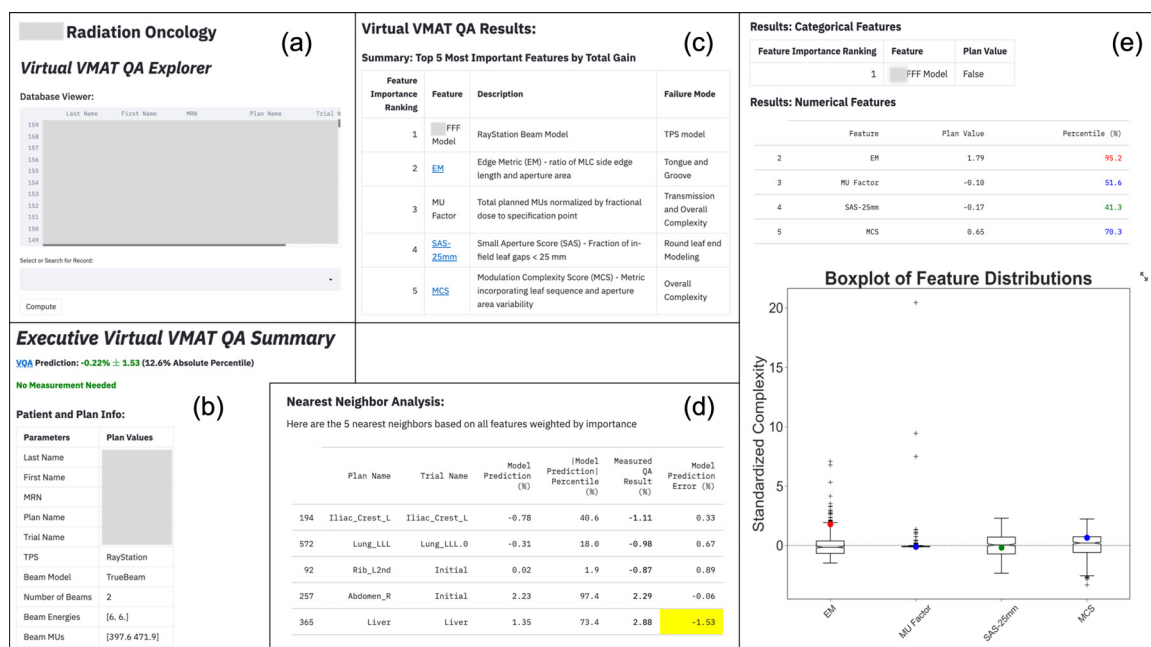


Fig. 1. Screenshots of the virtual quality assurance (VQA) web-based application deployed in our clinic. The user can search for a specific plan in the database (a), before computing the VQA prediction (b) along with additional plan analysis including feature importance (c), nearest neighbor analysis (d), and sample feature analysis relative to the model data set distributions (e). *Abbreviations:* QA = quality assurance; VMAT = volumetric modulated arc therapy.

Table 2 Summary of most important features as measured by different metrics by the XGBoost model

Importance rank	Coverage	Gain	Weight
1	wtdJAA-5	FFF beam model	MU factor
2	maxSAS-13mm	EM	EM
3	maxSAS-14mm	SAS-25mm	MCS
4	Collimator angle	FAOC-15mm	Dose rate
5	SAS-10mm	SAS-10mm	FFF beam model
6	LT-4	maxSAS-20mm	wtdJP-5
7	maxSAS-15mm	FAOC-30mm	wtdFAOC-10mm
8	wtdLG-5	avgXIJ	JAA-2
9	wtdJAA-4	FAOC-25mm	JPI
10	AA-3	SAS-35mm	FAOC-35mm

Refer to [Table 1](#) for detailed descriptions of features.

Abbreviations: AA-N = Nth moment of aperture area distribution; avgXIJ = average position of X1 jaw; EM = edge metric; FAOC-Xmm = fraction of area outside circle with X mm radius; JAA-N = Nth moment of jaw aperture area distribution; JP-N = Nth moment of jaw position distribution; JPI = jaw plan irregularity; LG-N = Nth moment of leaf gap distribution; LT-N = Nth moment of leaf travel distribution; maxSAS-Xmm = the maximum small aperture score at X mm over all segments; MCS = modulation complexity score; MU = monitor unit; SAS-Xmm = small aperture score at X mm; wtd = weighted by segment MUs.

and weight importance metrics. Briefly, coverage quantifies the relative number of samples related to a given feature, gain quantifies the relative contribution of the corresponding feature to the model, and weight quantifies the relative number of occurrences of a given feature in the trees of the model. Notably, the binary FFF Beam Model feature, scored highest in terms of gain, whereas numerical features relating to MLC and jaw positions and sequences (eg, the Edge Metric, Modulation Complexity Score, and Small Aperture Scores) comprised most of the other important features.

Initial model tuning and testing using 10-fold cross validation resulted in average training and testing mean absolute errors (MAE) of $0.73 \pm 0.14\%$ and $1.10 \pm 0.17\%$. Note that MAE and other prediction error metrics are reported as percentages to reflect the differences between predicted and true ion chamber disagreements, which are represented as and have units of percent dose differences. Twenty-four testing samples with prediction errors of greater than 3% were remeasured to account for setup errors such as selecting a nonideal measurement point near a high-dose gradient, which indeed was the culprit for a majority of these cases. After remeasurement, the model was retrained and reoptimized, where a maximum tree depth of 3 and a learning rate of 0.01 were selected as best performing hyperparameters. Subsequently all testing samples were predicted within 3% and the average testing MAE improved to $1.01 \pm 0.08\%$. [Figure 2](#) shows training and testing results from an independent 90/10 training/testing validation of the tuned model. A 90/10 validation split was selected to maximize performance on this small data set. Finally, the optimized model was trained on the entire data set before being deployed into the clinical workflow via a web-based VQA application for prospective validation. Early stopping was used, with 396 boosting rounds determined to give the best

results represented by MAE in the testing set during model validation.

ROC analysis was performed using the final model (tuned model trained on all available retrospective data before deployment) to generate predictions for all retrospective data (ie, combined training and testing data) to evaluate the model's baseline performance and to evaluate varying prediction thresholds for requiring a physical IC PSQA measurement ([Fig. 3](#)). Varying the clinical action level threshold from 1% to 4%, the area under the curve (AUC) steadily increased from 0.81 to 0.97. Using a 3% clinical threshold, the largest absolute VQA prediction threshold that yielded 100% sensitivity was 1.10% and also resulted in 65.6% specificity. In other words, a VQA prediction outside of $\pm 1.10\%$ would identify all plans that would result in IC measurements outside of $\pm 3\%$.

Prospective clinical validation

Over the course of 3 months, 445 VMAT plans were measured at our institution, taking an average of 19.0 minutes per plan for QA (based on a 1-month log). The mean absolute error of these 445 ion chamber measurements was 1.38% with a standard deviation of 0.94%. Initially, all but 14 PSQA cases were predicted to within 3%. Those 14 plans were subsequently remeasured with a more carefully selected point of measurement, after which each of the 14 PSQA cases were found to be within 3% of their VQA predictions. The VQA predictions resulted in a mean absolute error of $1.08 \pm 0.77\%$, with a maximum absolute error of 2.98% and $R^2 = 0.07$ ([Fig. 4](#)). A conservative VQA prediction decision threshold of 1%, (ie, plans with absolute predictions of less than 1% would not require IC

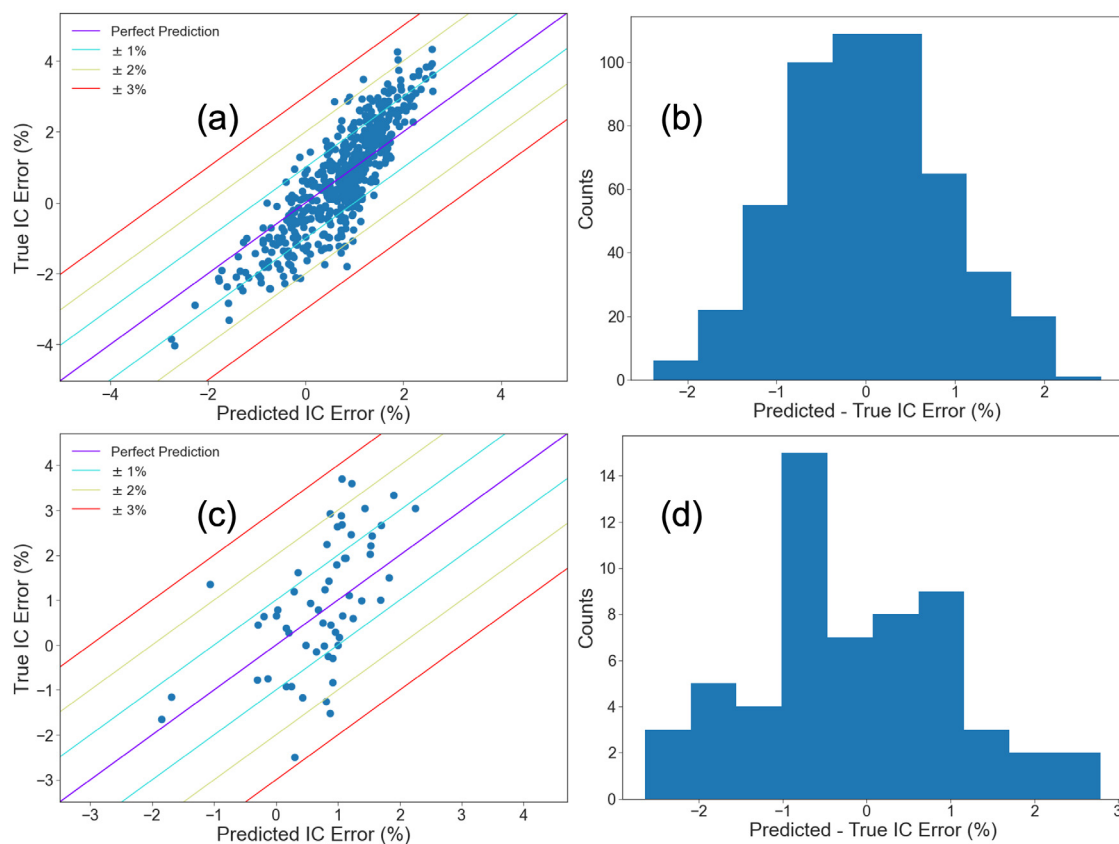


Fig. 2. Training (a, b) and testing (c, d) results from a 90/10 validation split on the tuned model. Training and testing mean absolute errors and R^2 values were 0.70% and 0.66 and 1.01% and 0.31, respectively. *Abbreviation:* IC = ion chamber.

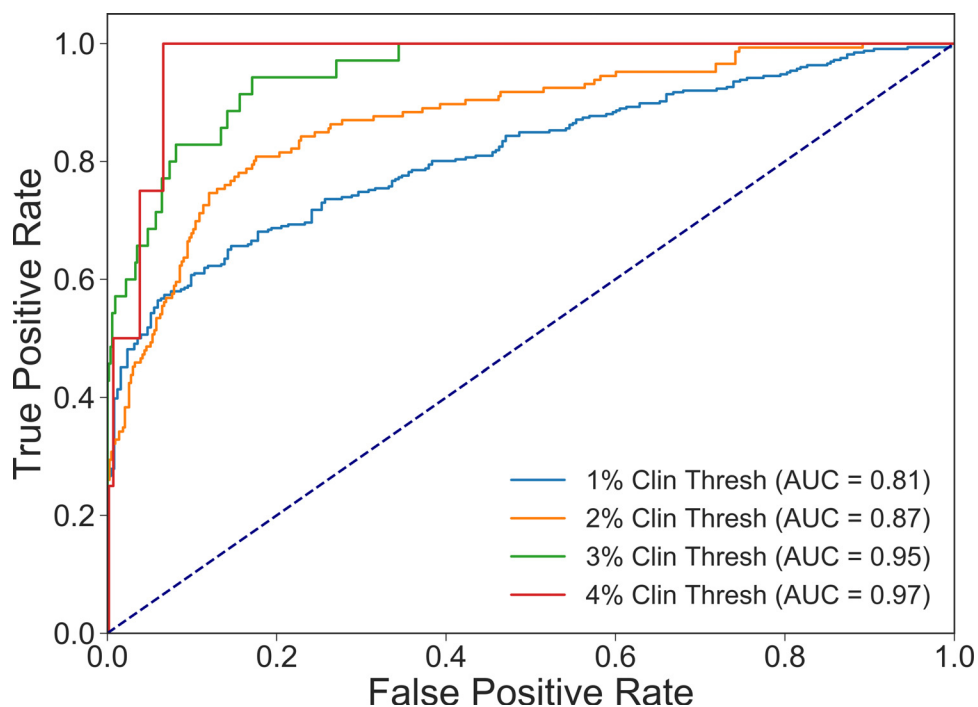


Fig. 3. Receiver operating characteristic curve for deployed model on retrospective data set with varying clinical decision thresholds of 1%, 2%, 3%, and 4%. *Abbreviation:* AUC = area under the curve.

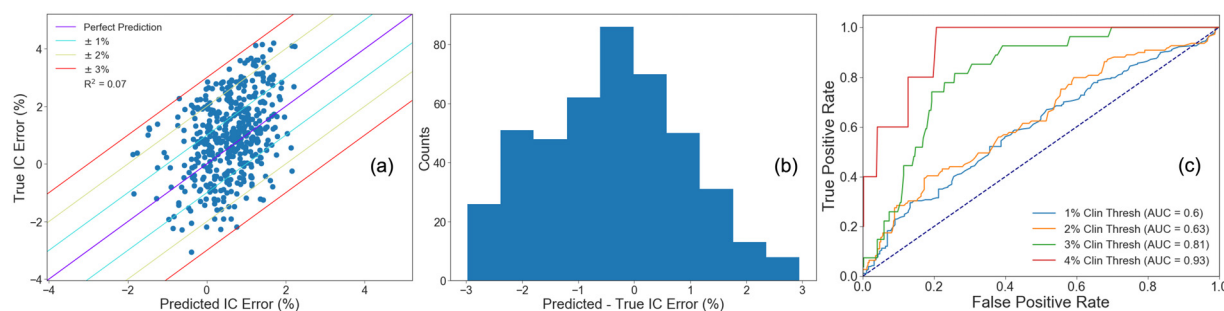


Fig. 4. Results from prospective clinical validation. All 445 volumetric modulated arc therapy patient-specific quality assurance measurements performed at our multisite clinic over the course of 3 months were predicted within 3% (a), resulting in a mean absolute error of $1.10 \pm 0.82\%$ (b). (c) Receiver operating characteristic curve for deployed model on prospective data set with varying clinical decision thresholds of 1%, 2%, 3%, and 4%. *Abbreviations:* AUC = area under the curve; IC = ion chamber.

measurements) was used which would result in an 81.5% sensitivity, 72.5% specificity, and 0.81 AUC using a 3% clinical threshold, and a 100% sensitivity and 70% specificity using a 4% clinical threshold. This 1% decision threshold, which would identify all plans that would result in measurement errors of greater than 4% error, translates to a 69.2% (approximately 32.5 hours) reduction in QA workload per month (ie, 69.2% of cases had VQA predictions between $\pm 1\%$). Further ROC analysis on this prospective data set is shown in Figure 4c, with AUC increasing from 0.6 to 0.93 when varying the clinical action level from 1% to 4%, respectively.

Discussion

In this work, a ML model was developed to predict IC disagreements for VMAT PSQA and implemented within an existing clinical workflow for prospective validation. Specifically, an XGBoost model based on previous VMAT treatment plans at our institution was integrated within a custom web-based application designed for clinical use by our QA team. The predictive model was able to predict all 445 prospective clinical VMAT PSQA results to within 3% accuracy. Using the underlying model as a classification clinical decision-support tool, a conservative 1% VQA threshold would yield a 69.2% reduction in QA workload, saving over 32 hours of QA time per month. This threshold also ensures measurement of all plans that may result in clinical PSQA outcomes of greater than 4%, which was confirmed in the retrospective and prospective analysis. However, this decision threshold can be adjusted to a given institution's data, action limits, and risk tolerance. Overall, we have demonstrated the feasibility of clinically implementing a predictive VMAT PSQA decision-support system and observed it to be efficient and effective in a prospective clinical analysis.

To our knowledge, this is the first study presenting a predictive model for calibrated ion chamber disagreements, considered to be among the most accurate and effective for

patient-specific absolute dose verification,^{16,23,48-51} for PSQA from solely planned parameters. Additionally, this work successfully implemented this predictive model within a user-friendly web application deployed for clinical use at a multisite radiation oncology department, where the VQA model and workflow was prospectively validated on a large data set. This represents a substantial step forward in the effort for maximizing efficiency and safety of the current standards of PSQA. Supplementing these model predictions on the VQA application are feature importance analyses and visualizations such as the numerical results of each important feature and a feature-specific boxplot showing the statistical distribution of this plan in relation to the training plan database. These features can be used to aid physicist interpretation of the prediction results in ultimately determining the need for a physical measurement and the plan approval for treatment.

Although the fundamental goal of ensuring the planned dose is delivered as intended within an acceptable tolerance is common to all PSQA methods, there are currently many different implementations. Measurement-based PSQA has long been considered the standard-of-care, while other methods have evolved over the years that include EPID-, software-, and log file-based PSQA.⁵²⁻⁵⁵ Each individual technique comes with its own set of advantages and disadvantages relating to effectiveness, efficiency, cost, and consistency that may lead a particular clinic or institution to prefer one method over the other depending on specific goals and needs. And while the value of PSQA in general has recently come into question, measurement-based methods are still widely used and remain the foundation of most PSQA programs.¹⁶

Previous investigators have researched myriad techniques to predict QA outcomes for different measurement and analysis methods.^{26,56,57} Of particular note, a recent study by Yang et al³⁶ presented a framework for developing, testing, and clinically implementing an Autoencoder-based Classification-Regression VQA model. Although this study collected a large, multi-institutional retrospective data set of 1835 VMAT plans for model development and testing,

previous work has shown similar predictive VQA models to plateau in testing performance at 200 samples.²⁹ In addition to this model development, an end-to-end testing framework on 20 prospectively collected plans was proposed in addition to only 50 implemented plans. This combined prospective data set of 70 samples is less than all but one of the testing sets referenced in the multiple testing scenarios and is less than 16% of the prospective data set presented here, which is a distinguishing aspect of this work. Furthermore, the present results build on this existing literature by developing a model for IC disagreements that can be generalized to any institution and prospectively validating its performance within the clinical workflow over 3 months.

Another important advantage of the present method over most existing software-based methods is the predelivery mechanical feature analysis supplementing the PSQA outcome prediction. These features are extracted and computed directly from the clinical DICOM RT Plan file (ie, no machine log-files are required) and are compared with distributions from a retrospective database of historical plans to inform the physicist how “complex” the given plan is relative to previous clinical data. Furthermore, this information could be extended to the dosimetry team during planning to indicate a given plan’s complexity and whether it was likely to have higher PSQA error rates *a priori*.

The catalog of literature has established the ability to understand the parameters that affect disagreements between calculated and measured doses and accurately identify plans that are mostly likely to result in high levels of disagreements, including how those parameters may depend on the site treated.⁵⁸ This study further complements these works with the development and testing of an XGBoost regression model. The novel and innovative aspect of this work is the demonstration of a complete and seamless integration of such a VQA model into the clinical workflow and prospectively evaluating its performance and potential efficiency gains in QA workload and resources. Although measurement-based PSQA does detect clinical errors, very few plans fail conventional evaluation metrics which facilitates VQA to safely reduce the number of required measurements. It is important to note that a VQA prediction model alone is not sufficient for robust measurement-free PSQA, especially one that predicts disagreements for a single point or small-volume measurement. However, given the questionable effectiveness and value of standard measurement-based PSQA methods shown in the literature, combined with the proven ability for independent dose calculation software to outperform measurement-based PSQA in detecting high severity and clinically relevant errors, a VQA model paired with such an independent second check can provide an equivalently effective and more efficient VMAT PSQA program. These 2 verification modalities complement each other in that secondary dose calculation methods lack the measurement error information VQA prediction provide. Additionally, being able to provide *a priori* knowledge of a given plan’s likelihood of failing PSQA upstream during the planning process decreases the chances of having to

replan after a failing PSQA measurement, which could help alleviate a highly disruptive scenario for staff and patients. Furthermore, the VQA platform is the only method that can inform the planner of a possible delivery limitation without delivering the plan. Building this synergy by connecting treatment planning and delivery QA processes offers an avenue for comprehensive adaptive radiation therapy, and provides avenues of future research into this potential paradigm shift in the radiation therapy planning workflow.

A limitation in a software-based PSQA protocol is the lack of verification of correct data transfer from the record and verify system to the linac sequencer, which is required to ensure data corruption has not caused a major error in a given plan.⁵⁹ A possible solution to verifying accurate data transfer and deliverability is log-file analysis. There is commercially available software, including M3D, providing independent recalculations of dose using linac log-files, which is currently used for assessing VMAT PSQA at our department. Therefore, a “dry run” would still be needed for each plan to generate log files for each patient plan until a more integrative solution can be formed with VQA, which is an ongoing area of research being investigated by our group. Furthermore, we recommend continuing to use such software-based independent recalculation QA tools in combination with any VQA prediction system, including the one presented in this work. Our institution has also decided to continue measuring all SBRT plans, but this is a reflection on the risk tolerance rather than the performance of the model. This proposed VQA framework is undergoing internal new process review before clinical operation.

Another limitation of this study is how inherent errors in the PSQA measurement process affect the resulting VQA predictive model. As seen when remeasurements were performed in both the training and prospective validation stages, PSQA outcomes for a single IC dose measurement is highly dependent on setup and point selection within the planned dose distribution. Because the XGBoost model was designed to predict QA outcomes assuming measurement conditions met clinical standards, any points found to or suspected of not meeting these standards should be remeasured. Furthermore, although the present VQA model yields PSQA measurement predictions for any measurement point meeting the clinical criteria of our institution (ie, not for a specifically defined point), incorporating location-specific information found outside the DICOM RT Plan file could potentially improve predictive performance and is an avenue for future research. However, given the advantage of IC measurements yielding absolute dose in contrast to relative measurements like the majority of PSQA measurement systems across institutions, IC measurements still typically have smaller uncertainties than film or detector array measurements. Additionally, although there were no “failing” (>5%) PSQA results in the training data set, there were also no “failing” PSQA results over the 3 months of prospective clinical validation. This underscores the value in moving toward identifying plans that are less likely to result in high levels of PSQA disagreements and not require physical

measurements so that resources and focus can be more effectively allocated to the most complex cases.

In addition to measurement-based pretreatment PSQA being one of the least effective quality checks in the entire radiation therapy process according to Ford et al,²¹ VMAT PSQA rarely results in failing plans given modern TPS modeling and delivery systems. Furthermore, these results offer numerous potential paradigm-shifting options for implementing VQA safely and effectively. In addition to the proposal outlined previously, VQA could be used to identify subset of VMAT plans that sample the historical spectrum of complexity and high levels of measurement disagreements and deliver these subset of plans as part of a daily, weekly, monthly, and annual quality assurance tests in accordance to the principles of Task Groups and Practice Guidelines outlined by the American Association of Physicists in Medicine.^{8,9,60} A more comprehensive failure modes and effects analysis of any specific clinical implementation of VQA would therefore be warranted to robustly assess potential sources of error.

Conclusions

This study is the first to successfully implement a VQA model into an existing clinical workflow for purposes of reducing the number of physical measurements required for PSQA. Additionally, this work presents the first model for predicting calibrated ion chamber measurements for PSQA, which is among the most accurate methods for measuring absolute dose in PSQA according to standard clinical reference dosimetry protocols. The tuned model was able to predict all samples to within 3% accuracy, performing similarly on the prospective clinical samples. Using a 1% VQA prediction threshold, upwards of 70% reduction in QA workload can be achieved while maintaining safety and quality. Furthermore, this work represents the next step toward measurement-free PSQA in the clinic as radiation oncology departments around the globe search for value-based methods to increase efficiency.

References

- Otto K. Volumetric modulated arc therapy: IMRT in a single gantry arc. *Med Phys* 2008;35:310–317.
- Teoh M, Clark CH, Wood K, Whitaker S, Nisbet A. Volumetric modulated arc therapy: A review of current literature and clinical use in practice. *Br J Radiol* 2011;84:967–996.
- Davidson MT, Blake SJ, Batchelar DL, Cheung P, Mah K. Assessing the role of volumetric modulated arc therapy (VMAT) relative to IMRT and helical tomotherapy in the management of localized, locally advanced, and post-operative prostate cancer. *Int J Radiat Oncol Biol Phys* 2011;80:1550–1558.
- Deng Z, Shen L, Zheng X, et al. Dosimetric advantage of volumetric modulated arc therapy in the treatment of intraocular cancer. *Radiat Oncol* 2017;12:83.
- Nguyen K, Cummings D, Lanza VC, et al. A dosimetric comparative study: Volumetric modulated arc therapy vs intensity-modulated radiation therapy in the treatment of nasal cavity carcinomas. *Med Dosim* 2013;38:225–232.
- Quan EM, Li X, Li Y, et al. A comprehensive comparison of IMRT and VMAT plan quality for prostate cancer treatment. *Int J Radiat Oncol Biol Phys* 2012;83:1169–1178.
- Tsai CL, Wu JK, Chao HL, Tsai YC, Cheng JC. Treatment and dosimetric advantages between VMAT, IMRT, and helical tomotherapy in prostate cancer. *Med Dosim* 2011;36:264–271.
- Klein EE, Hanley J, Bayouth J, et al. Task Group 142 report: Quality assurance of medical accelerators. *Med Phys* 2009;36:4197–4212.
- Smith K, Balter P, Duhon J, et al. AAPM medical physics practice guideline 8.a: Linear accelerator performance tests. *J Appl Clin Med Phys* 2017;18:23–39.
- Nelms BE, Robinson G, Markham J, et al. Variation in external beam treatment plan quality: An inter-institutional study of planners and planning systems. *Pract Radiat Oncol* 2012;2:296–305.
- Court L, Wagar M, Berbeco R, et al. Evaluation of the interplay effect when using RapidArc to treat targets moving in the craniocaudal or right-left direction. *Med Phys* 2010;37:4–11.
- Ezzell GA, Burmeister JW, Dogan N, et al. IMRT commissioning: Multiple institution planning and dosimetry comparisons: A report from AAPM Task Group 119. *Med Phys* 2009;36:5359–5373.
- Galvin JM, Ezzell G, Eisbrauch A, et al. Implementing IMRT in clinical practice: A joint document of the American Society for Therapeutic Radiology and Oncology and the American Association of Physicists in Medicine. *Int J Radiat Oncol Biol Phys* 2004;58:1616–1634.
- Yu CX, Jaffray DA, Wong JW. The effects of intra-fraction organ motion on the delivery of dynamic intensity modulation. *Phys Med Biol* 1998;43:91–104.
- Das JJ, Ding GX, Ahnesjo A. Small fields: Nonequilibrium radiation dosimetry. *Med Phys* 2008;35:206–215.
- Miften M, Olch A, Mihailidis D, et al. Tolerance limits and methodologies for IMRT measurement-based verification QA: Recommendations of AAPM Task Group No. 218. *Med Phys* 2018;45:e53–e83.
- Handsfield LL, Jones R, Wilson DD, Siebers JV, Read PW, Chen Q. Phantomless patient-specific TomoTherapy QA via delivery performance monitoring and a secondary Monte Carlo dose calculation. *Med Phys* 2014;41:101703.
- Kry SF, Glenn MC, Peterson CB, et al. Independent recalculation outperforms traditional measurement-based IMRT QA methods in detecting unacceptable plans. *Med Phys* 2019;46:3700–3708.
- Siochi RA, Molineu A, Orton CG. Point/counterpoint. Patient-specific QA for IMRT should be performed using software rather than hardware methods. *Med Phys* 2013;40:070601.
- Pan Y, Yang R, Zhang S, et al. National survey of patient specific IMRT quality assurance in China. *Radiat Oncol* 2019;14:69.
- Ford EC, Terezakis S, Souranis A, Harris K, Gay H, Mutic S. Quality control quantification (QCQ): A tool to measure the value of quality control checks in radiation oncology. *Int J Radiat Oncol Biol Phys* 2012;84:e263–e269.
- Nelms BE, Zhen H, WA Tome. Per-beam, planar IMRT QA passing rates do not predict clinically relevant patient dose errors. *Med Phys* 2011;38:1037–1044.
- Kry SF, Molineu A, Kerns JR, et al. Institutional patient-specific IMRT QA does not predict unacceptable plan delivery. *Int J Radiat Oncol Biol Phys* 2014;90:1195–1201.
- Interian Y, Rideout V, Kearney VP, et al. Deep nets vs expert designed features in medical physics: An IMRT QA case study. *Med Phys* 2018;45:2672–2680.
- Li J, Wang L, Zhang X, et al. Machine learning for patient-specific quality assurance of VMAT: Prediction and classification accuracy. *Int J Radiat Oncol Biol Phys* 2019;105:893–902.
- Ono T, Hirashima H, Iramina H, et al. Prediction of dosimetric accuracy for VMAT plans using plan complexity parameters via machine learning. *Med Phys* 2019;46:3823–3832.
- Tomori S, Kadota N, Takayama Y, et al. A deep learning-based prediction model for gamma evaluation in patient-specific quality assurance. *Med Phys* 2018;45:4055–4065.

28. Valdes G, Chan MF, Lim SB, Scheuermann R, et al. IMRT QA using machine learning: A multi-institutional validation. *J Appl Clin Med Phys* 2017;18:279–284.
29. Valdes G, Scheuermann R, Hung CY, Olszanski A, et al. A mathematical framework for virtual IMRT QA using machine learning. *Med Phys* 2016;43:4323.
30. Wall PDH, Fontenot JD. Application and comparison of machine learning models for predicting quality assurance outcomes in radiation therapy treatment planning. *Informatics in Medicine Unlocked* 2020;18:100292.
31. Lizar JC, Yaly CC, Colello Bruno A, Viani GA, Pavoni JF. Patient-specific IMRT QA verification using machine learning and gamma radiomics. *Phys Med* 2021;82:100–108.
32. Tomori S, Kadoya N, Kajikawa T, et al. Systematic method for a deep learning-based prediction model for gamma evaluation in patient-specific quality assurance of volumetric modulated arc therapy. *Med Phys* 2021;48:1003–1018.
33. Wang L, Li J, Zhang S, et al. Multi-task autoencoder based classification-regression model for patient-specific VMAT QA. *Phys Med Biol* 2020;65:235023.
34. Potter NJ, Mund K, Andreozzi JM, Li JG, Liu C, Yan G. Error detection and classification in patient-specific IMRT QA with dual neural networks. *Med Phys* 2020;47:4711–4720.
35. Hirashima H, Ono T, Nakamura M, et al. Improvement of prediction and classification performance for gamma passing rate by using plan complexity and dosimetrics features. *Radiother Oncol* 2020;153:250–257.
36. Yang R, Yang X, Wang L, et al. Commissioning and clinical implementation of an Autoencoder based Classification-Regression model for VMAT patient-specific QA in a multi-institution scenario. *Radiother Oncol* 2021;161:230–240.
37. Valdes G, Adamson J, Cai J. Artificial intelligence for prediction of measurement-based patient-specific quality assurance is ready for prime time. *Med Phys* 2021;48:2701–2704.
38. Hasse JK, Scholey J, Ziemer BP, et al. Use of receiver operating curve analysis and machine learning with an independent dose calculation system reduces the number of physical dose measurements required for patient-specific quality assurance. *Int J Radiat Oncol Biol Phys* 2021;109:1086–1095.
39. Almond PR, Biggs PJ, Coursey BM, et al. AAPM's TG-51 protocol for clinical reference dosimetry of high-energy photon and electron beams. *Med Phys* 1999;26:1847–1870.
40. Crowe SB, Kairn T, Kenny J, et al. Treatment plan complexity metrics for predicting IMRT pre-treatment quality assurance results. *Australas Phys Eng Sci Med* 2014;37:475–482.
41. Du W, Cho SH, Zhang X, Hoffman KE, Kudchadker RJ. Quantification of beam complexity in intensity-modulated radiation therapy treatment plans. *Med Phys* 2014;41:021716.
42. Masi L, Doro R, Favuzza V, Cipressi S, Livi L. Impact of plan parameters on the dosimetric accuracy of volumetric modulated arc therapy. *Med Phys* 2013;40:071718.
43. McNiven AL, Sharpe MB, Purdie TG. A new metric for assessing IMRT modulation complexity and plan deliverability. *Med Phys* 2010;37:505–515.
44. Younge KC, Matuszak MM, Moran JM, McShan DL, Fraass BA, Roberts DA. Penalization of aperture complexity in inversely planned volumetric modulated arc therapy. *Med Phys* 2012;39:7160–7170.
45. XGBoost. XGBoost documentation. Available at: <https://xgboost.readthedocs.io/en/latest/index.html>. Accessed June 30, 2022.
46. Chen T, Guestrin C. XGBoost: A scalable tree boosting system. *Paper presented at: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. 2016; Association for Computing Machinery; New York, NY and San Francisco, CA, USA.
47. Streamlit. Streamlit documentation. Available at: <https://docs.streamlit.io/en/stable/index.html>. Accessed June 30, 2022.
48. McKenzie EM, Balter PA, Stingo FC, Jones J, Followill DS, Kry SF. Toward optimizing patient-specific IMRT QA techniques in the accurate detection of dosimetrically acceptable and unacceptable patient plans. *Med Phys* 2014;41:121702.
49. Dong L, Antolak J, Salehpour M, et al. Patient-specific point dose measurement for IMRT monitor unit verification. *Int J Radiat Oncol Biol Phys* 2003;56:867–877.
50. Kruse JJ. On the insensitivity of single field planar dosimetry to IMRT inaccuracies. *Med Phys* 2010;37:2516–2524.
51. Pulliam KB, Followill D, Court L, et al. A six-year review of more than 13,000 patient-specific IMRT QA results from 13 different treatment sites. *J Appl Clin Med Phys* 2014;15:196–206.
52. Howell RM, Smith IP, Jarrio CS. Establishing action levels for EPID-based QA for IMRT. *J Appl Clin Med Phys* 2008;9:16–25.
53. Defoor DL, Stathakis S, Roring JE, et al. Investigation of error detection capabilities of phantom, EPID and MLC log file based IMRT QA methods. *J Appl Clin Med Phys* 2017;18:172–179.
54. Olch AJ. Evaluation of the accuracy of 3DVH software estimates of dose to virtual ion chamber and film in composite IMRT QA. *Med Phys* 2012;39:81–86.
55. Sun B, Rangaraj D, Boddu S, et al. Evaluation of the efficiency and effectiveness of independent dose calculation followed by machine log file analysis against conventional measurement based IMRT QA. *J Appl Clin Med Phys* 2012;13:140–154.
56. Chan MF, Witztum A, Valdes G. Integration of AI and machine learning in radiotherapy QA. *Front Artif Intell* 2020;3:577620.
57. Granville DA, Sutherland JG, Belec JG, La Russa DJ. Predicting VMAT patient-specific QA results using a support vector classifier trained on treatment plan characteristics and linac QC metrics. *Phys Med Biol* 2019;64:095017.
58. Li J, Zhang X, Li J, et al. Impact of delivery characteristics on dose delivery accuracy of volumetric modulated arc therapy for different treatment sites. *J Radiat Res (Tokyo)* 2019;60:603–611.
59. Bogdanich W. Radiation offers new cures, and ways to do harm. *New York Times* 2010;23:A1.
60. Kutcher GJ, Coia L, Gillin M, et al. Comprehensive QA for radiation oncology: Report of AAPM Radiation Therapy Committee Task Group 40. *Med Phys* 1994;21:581–618.